

Customer Shopping Behaviour
CCDS 321 – Data Mining
Milestone 3

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Summary Statistical

The table below presents Descriptive measures for each attribute in the customer dataset. However, these statistical calculations help to understand the visualizations. These statistical summaries are measured by mean, median, and standard deviation. Meanwhile, variance hasn't been displayed in SAS, but it was calculated manually.

Filter		Descriptive Measures						Metadata Measures	Data Quality Measures
#	Name	Distinct Values	Mean	Median	Minimum	Maximum	Standard Deviation		
3	Customer ID	4,868	2466.609	2434.5	1	5099	1452.176		
4	Age	53	44.14544	44	18	70	15.27034		
8	Purchase Amount (USD)	494	124.9901	70	10.12	1499.76	224.784		
12	Review Rating	29	3.677002	3.668195	1	5	0.80238		
16	Previous Purchases	52	25.25218	25.2219	0	50	13.76855		
1	Frequency of Purchases	5	--	--	--	--	--		
2	Season	4	--	--	--	--	--		
5	Gender	3	--	--	--	--	--		
6	Item Purchased	30	--	--	--	--	--		
7	Category	5	--	--	--	--	--		

Variance calculations:

Age: 233.18

Purchase Amount: 50527.86

Review Rating:0.6438

Previous Purchases:189.57

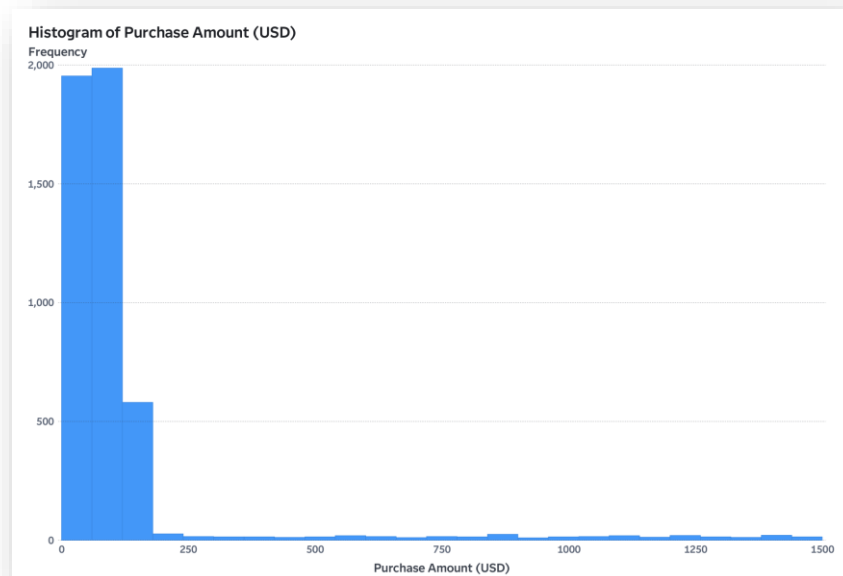
Insight: The variance calculations show the distribution of the numeric attributes. In addition, the variance value in **Purchase Amount** is (50527.86), which makes it the largest variance value from the remaining attributes. Meanwhile, **Review Rating** is (0.6438), which shows the smallest variance, indicating that the rating is balanced across customers. Also, **Previous Purchases** is (189.57), and **Age** is (233.18), which have a similar variance value. However, the variance will affect the visualizations.

Visualization

1. Univariate

- **Purchase Amount (USD)**

The histogram shows the **purchase amount (USD)** that customers spend.



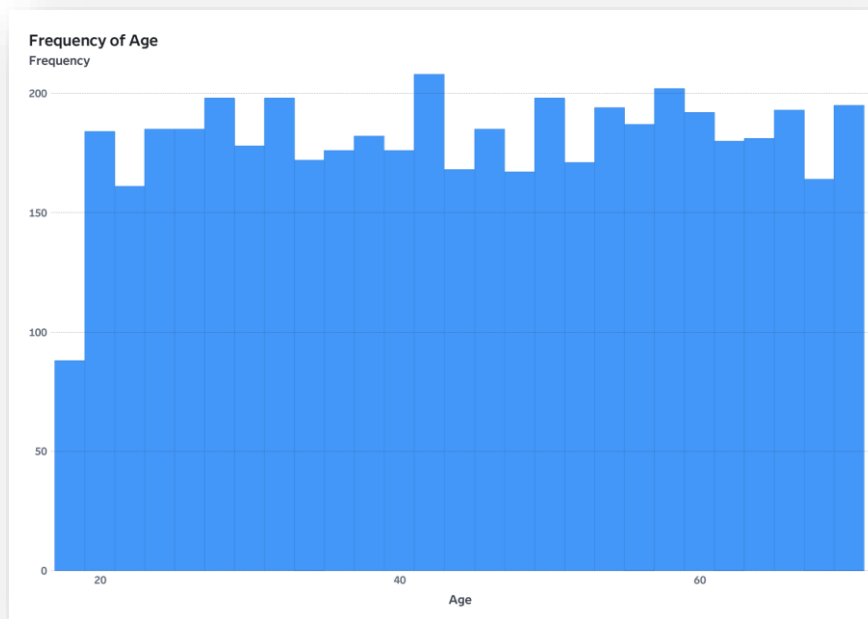
Key observation:

- The histogram plot has a **right-skewed** distribution, which explains that the purchases are under 250\$.
- There are a few customers whose purchase cost **1500\$**. Therefore, it explains the high variance in summary statistics.
- The mean is (124.99\$) while the median is (70\$). As a result, high numbers make the average value higher.

Insight: Most customers tend to spend less. There is a gap between the mean and median, caused by high spenders, that affects the frequency of purchase amounts.

- **Age**

The histogram shows the **age** of customers



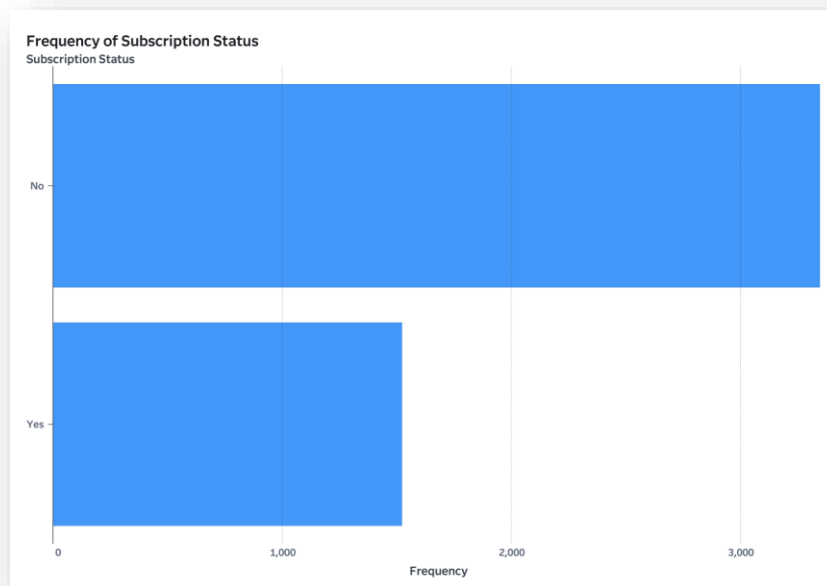
Key observations:

- The histogram shows the distribution appears balanced across all ages from **17 to 71 years**.
- The mean is (44.14), and the median is (44), which exactly equals. Then the balanced distribution is **symmetric**.
- There is **no dominant age** in the distribution, which means the company does not target a specific age.

Insight: The company provides for customers of all ages and it's not limited to a specific age group. As a result, the distribution is symmetric.

- **Subscription Status**

The bar chart shows the **Subscription Status** of customers



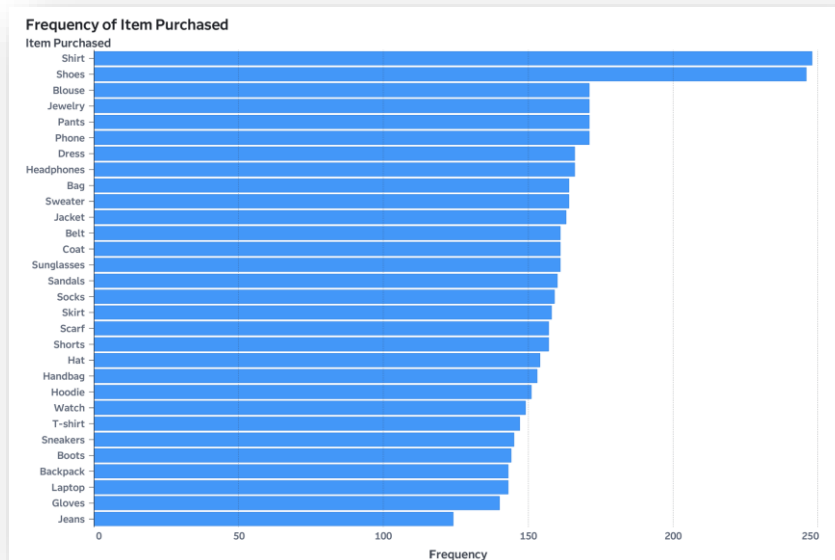
Key observations:

- Most **non-subscriber** customers have a frequency of 3,346.
- A few **subscriber** customers have a smaller frequency of 1,522.
- Non-subscribers are double the number of subscribers, which indicates a huge gap.

Insight: The bar chart shows that most customers are not subscribers, which is advantageous to convince customers to subscribe, due to their purchase amount in a bivariate bar chart.

- **Item Purchased**

The bar chart shows that customers purchased items



Key observations:

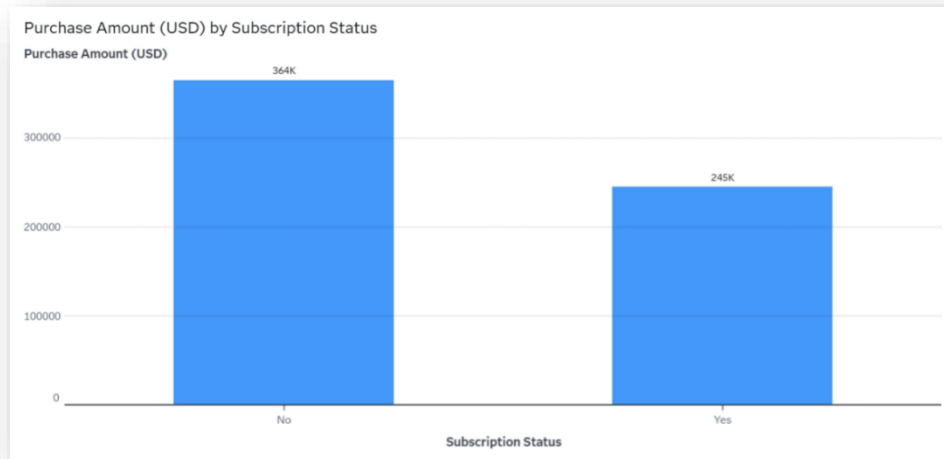
- The two most purchased items are shirts and shoes, with frequent purchases of 248 and 246.
- The remaining items purchased are frequently equal, for example, blouse, bag, hat, and boots with frequent from 171 to 140.
- The least purchased item is jeans with a frequency of 124.

Insight: The bar chart shows the frequency of all purchases. Therefore, clothes are the most purchased, jeans are the least purchased, and the remaining items have a balanced frequency.

2. Bivariate

- **Bar Chart Analysis**

The bar chart compares the total **Purchase Amount (USD)** between subscribers and non-subscribers.



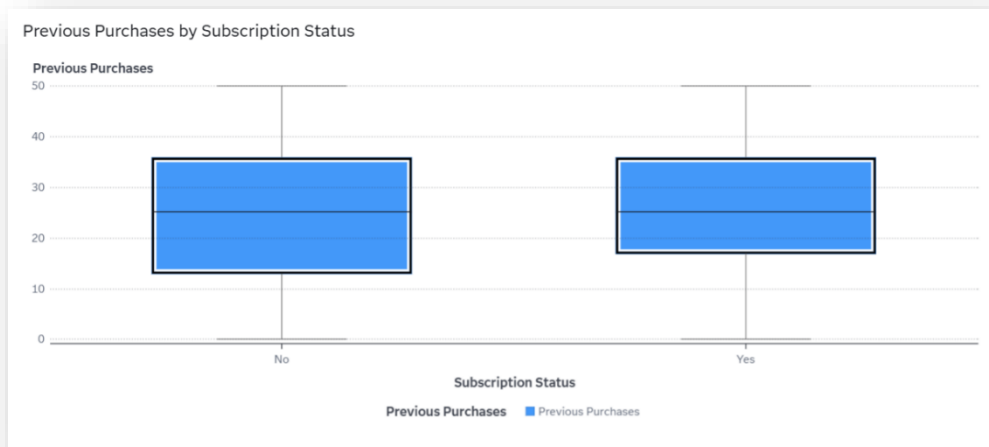
Key observations:

- Non-subscribers contribute a significantly higher total revenue, amounting to **approximately \$364K**.
- Subscribers account for a lower total spending, reaching **approximately \$245K**.
- The gap in spending shows that the **bulk of total sales** currently comes from the non-subscriber segment.

Insight: Based on the data, while subscribers are important, the majority of revenue comes from non-subscribers. This reveals a significant opportunity to increase subscription value by targeting these high-spending customers.

- **Box Plot Analysis**

The box plot displays the distribution of **Previous Purchases** based on **Subscription Status**.



Subscription Status ▲	Minimum	Lower Whisker	First Quartile	Average	Median	Third Quartile	Upper Whisker	Maximum	Std Dev	Count
Yes	0	0	17	25.925924002	25.221901377	36	50	50	12.981424887	1,522
No	0	0	13	24.945716893	25.221901377	36	50	50	14.103310483	3,346

Key observations:

- Both subscribed and non-subscribed customers show **identical shopping habits**, with a median of **approximately 25 purchases**.
- The range of transactions is the same for both groups, spreading from **0 to 50 purchases**.
- There is no significant difference in how often a customer buys, regardless of whether they are a subscriber or not.

Insight: Previous purchase frequency alone does not show a clear effect on subscription status. Customers in both groups interact with the brand at the same rate.

3. Advanced Graph

- **Bubble Plot**

The bubble plot displays the relationship between **Purchase Amount**, **Previous Purchases**, and **Review Rating**, colored by **Subscription Status**.



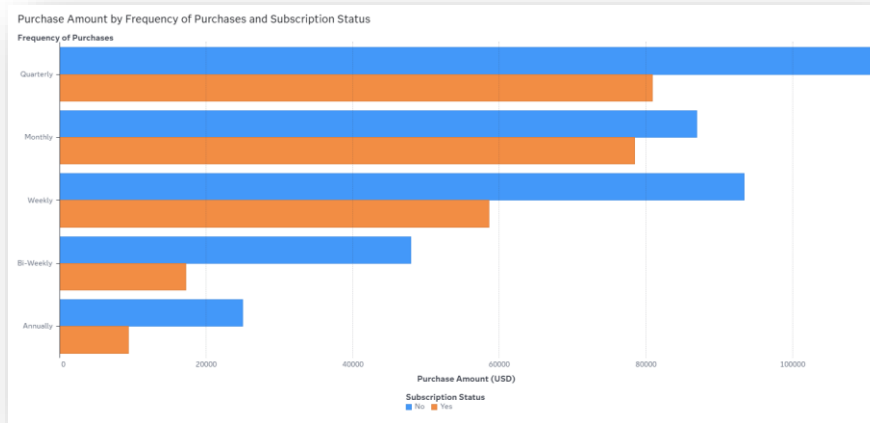
Key observations:

- Both subscribed and non-subscribed customers are distributed across different purchase amount ranges and previous purchase counts, with no clear separation between the two groups based on these variables alone.
- Most purchases are concentrated **below \$250**, regardless of subscription status.
- Customers with higher purchase amounts **above \$750** exist in both groups, suggesting that purchase amount alone does not determine subscription behavior.
- **Review Rating**, represented by bubble size, shows no significant variation between subscribed and non-subscribed customers.

Insight: Purchase Amount, Previous Purchases, and Review Rating alone do not show a clear effect on subscription status. A combination of multiple variables may be needed to better understand customer subscription behavior.

- **Grouped Bar Chart**

The bar chart shows the total **Purchase Amount** for each purchase frequency category, grouped by **Subscription Status**.

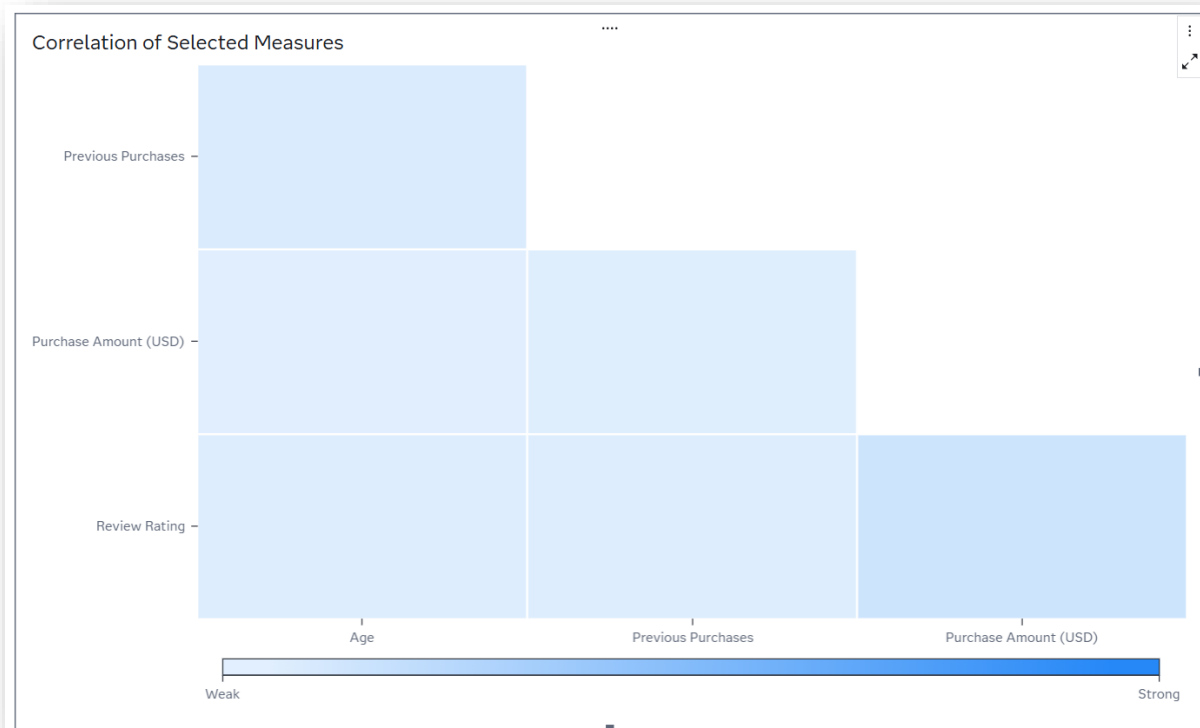


Key observations:

- **Quarterly** purchasers have the highest total spending in both groups, especially among non-subscribed customers.
- **Monthly** purchasers show close spending values between subscribed and non-subscribed customers.
- **Weekly** purchasers show higher spending among non-subscribed customers than subscribed customers.
- **Bi-Weekly** and **Annual** purchasers also show higher spending among non-subscribed customers.
- **Overall**, non-subscribed customers spend more across most purchase frequency categories.

Insight: Purchase amount alone does not seem to strongly affect subscription behavior. Even though non-subscribed customers spend more in most categories, subscription decisions may depend on other customer factors as well.

Correlation Matrix



X Axis	Y Axis	Correlation
Age	Review Rating	-0.0181
Age	Purchase Amount (USD)	0.0051
Age	Previous Purchases	0.0386
Previous Purchases	Review Rating	0.0168
Previous Purchases	Purchase Amount (USD)	-0.0095
Purchase Amount (USD)	Review Rating	-0.1055

Insight: The correlation matrix shows very weak relationships between the numeric variables. All correlation values are close to zero, which means there are no strong relationships between Age, Previous Purchases, Purchase Amount (USD), and Review Rating. These variables do not have a strong effect on each other.